Luca Rosu

Hardware Systems & Test Engineer | Hardware Systems & Data Acquisition | Cross-functional Program Execution

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SUMMARY

Hardware engineer with hands-on systems-integration experience across SpaceX, Tesla, and Apple — defining acceptance criteria and test infrastructure, resolving cross-disciplinary (mechanical/electrical/software) issues on electromechanical systems, and driving reliability improvements through root-cause failure analysis.

SKILLS

Robotics & Systems Integration: Robot/hardware bring-up & commissioning, system verification & validation testing, acceptance criteria & test-plan development, cross-disciplinary (mechanical/electrical/software) debug, actuators & electromechanical systems, PCB design (Altium), wiring harnesses, CAN bus, embedded systems (Raspberry Pi, ESP32)

Reliability & Test: Reliability/degradation testing, failure-mode analysis (FMEA), root-cause diagnosis, redundancy design, DAQ system design & build, precision soldering

Software & Automation: Python, C, MATLAB, Java, PyTorch; test automation & data pipelines; reinforcement learning (PPO, stable-baselines3, Gymnasium); Jira, Confluence

Manufacturing & Process: DFM/DFA, quality acceptance plans, vendor coordination & SLA accountability, CNC, welding, 3D printing

LEADERSHIP & PROJECTS

Purdue Electric Racing (SAE) — President ('24–'25), Drivetrain Lead ('23)
2021 – July 2025

- Led a 110-person team and $100K budget as President across all 7 sub-teams, leading the car to its first-ever completion of all competition events and a top-10 finish.
- Owned integration of the car's HV and LV electrical systems, driving cross-team coordination between the Battery, LV, and Drivetrain sub-teams to close communication and interface gaps between systems.
- Owned design and integration of all LV enclosures — packaging them into the chassis and battery pack alongside the Vehicle Dynamics team to ensure mechanical, electrical, and thermal fit across the full vehicle.

Autonomous Home Mapping Bot
Solo project, ongoing (refinement)

- Designed and built an autonomous rover with 2D LiDAR for mapping and localization and a camera for close-range obstacle detection and hazards outside the LiDAR's scan plane.
- Built the full autonomy stack solo: occupancy mapping plus a PPO navigation policy (stable-baselines3, Gymnasium API) trained with domain randomization for sim-to-real transfer; the rover now navigates autonomously.

EXPERIENCE

Tesla — HW Test Engineer, HV Devices
Palo Alto, CA · 2025 – Present

- Own system verification & validation strategy for the Tesla Semi electrical power takeoff (ePTO), helping resolving issues across PCBA-level electrical and mechanical abuse tests to hit an accelerated B-to-C sample timeline.
- Designed and built custom DAQ systems for ePTO verification and data collection, combining purpose-built hardware with Tesla's own ECUs for control; cuts manual data acquisition time by ~1 hour per unit.
- Owned actuator reliability testing: wrote scripts to cycle and monitor actuators across simulated field conditions, partnering with firmware and mechanical design teams to define test coverage and triage failures.
- Use AI coding tools, including Claude Code and Grok Build, for test script development and data analysis; cuts script iteration and debugging time ~50% versus coworkers, and helps introduce teammates to AI-assisted workflows.

SpaceX — Starship Mechanisms
Los Angeles, CA · May 2025 – Aug 2025

- Defined acceptance criteria, test procedures, and passing metrics for Starship V3 actuator stators, and built the failure-documentation system that established repeatable acceptance discipline for the line.
- Led failure-mode analysis on the actuator system and introduced redundancy strategies to improve reliability, later adopted by senior design leadership.
- Prototyped and validated a new stator manufacturing process with enhanced insulation that eliminated phase-to-phase arcing under high load, and developed assembly-line processes that improved step efficiency by 50%.

Apple — iPhone Display Engineering Program Mgmt
Cupertino, CA · May 2024 – Aug 2024

- Owned prototype-to-ramp build plans for iPhone display modules, coordinating across ME, EE, Optics, and Process teams.
- Drove vendor accountability across engineering and program teams — surfacing integration risks, escalating critical issues to vendors and leadership, managing allocation and scheduling during display bring-up.

Tesla — Hardware Test Engineer
Palo Alto, CA · Sep 2023 – Dec 2023

- Unblocked Cybertruck production ramp by diagnosing and resolving a high-voltage connector test failure.
- Designed, developed, and deployed degradation and reliability tests for pivotal components; identified failures and recommended design changes to mitigate them.

EDUCATION

Purdue University — B.S., Mechanical Engineering
2021 – 2025

Languages: English & Romanian (native), French. Citizenship: US / EU